Abdallah Yehya Soliman

Data Scientist | AI Engineer

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Summary

Computer Science graduate and AI Engineer with hands-on experience developing end-to-end AI systems across machine learning, deep learning, NLP, computer vision, and LLM applications. Experienced in building RAG pipelines, agentic AI workflows, natural-language-to-SQL systems, and AI-powered applications, with a strong foundation in Python, SQL, data processing, model evaluation, and backend development. Experienced in applying AI to practical problems through projects involving LLMs, retrieval, semantic search, database reasoning, and multi-model AI systems.

Experience

Business Intelligence Intern at Coca-Cola Egypt (Aug.2021-Sep.2021)

- Applied machine learning techniques to an AI-based analytics project, contributing to fraud detection improvements and reducing fraudulent activity by 5%.
- Built and trained an image classification model to recognize and categorize multiple product types, supporting automated data-driven workflows.
- Performed data preprocessing, feature extraction, and model evaluation to improve classification accuracy.
- Analyzed structured datasets and created interactive dashboards using Power BI, enabling stakeholders to make data-driven decisions.
- Collaborated with cross-functional teams to translate business requirements into machine learning solutions.

Web Developer (Freelance) Modern Broadcast Center (Jan.2025-Mar.2025)

- Developed and deployed a full-stack web application using Next.js, Prisma ORM, and PostgreSQL (Supabase), implementing product management and checkout workflows.
- Ensured responsive design and performance optimization for scalable deployment.

Backend Intern at Trans-IT (Sep.2023-Nov.2023)

- Strengthened problem-solving skills through backend development using data structures, algorithms, and OOP.
- Designed and managed relational databases using SQL Server, developing a database-driven Supermarket Management System covering inventory and sales operations.

Education

Bachelor’s degree in computer science (Ain-Shams University (2020-2024))

- Department: Computer Science
- GPA: 2.95

Graduation Project

Smart School System (Grade: A)

- Designed and implemented an end-to-end AI system applying machine learning and deep learning techniques for student monitoring and safety.
- Developed a face recognition pipeline using a Siamese network for accurate student identification.
- Integrated a liveness detection model to prevent faking attendance, enhancing system reliability.
- Built a violence detection model using sequence-based deep learning to monitor behavior and trigger alerts.
- Implemented a uniform compliance classification model using supervised learning techniques.

Academic & AI Projects

Agentic NL2SQL System

- Built an agentic AI system that translates natural-language questions into executable SQL queries across a multi-table relational database.
- Implemented metadata-driven schema routing and sub-schema selection to identify relevant tables, columns, and relationships before SQL generation.
- Designed SQL generation, validation, self-correction, and natural-language answer workflows for reliable database querying.
- Built the application with Next.js, TypeScript, Supabase/PostgreSQL, and an LLM-based backend workflow.

Arabic Sentiment Analysis (2023-2024)

- Built an end-to-end NLP pipeline including tokenization, lemmatization, stop-word removal, data augmentation, and emoji polarity labeling.
- Trained and evaluated RNN, LSTM, and Transformer-based models for sentiment classification.

News Group Classification (2022-2023)

- Built a text classification pipeline using NLTK and Scikit-learn, applying tokenization, lemmatization, stemming, stop-word removal, and data augmentation.
- Extracted TF-IDF features and trained Logistic Regression, SVM, and KNN models; evaluated Accuracy, Precision, Recall, and F1-score.

Movies Popularity Prediction (2022-2023)

- Developed a machine learning regression pipeline with preprocessing and feature engineering using Scikit-learn.
- Trained and compared Linear Regression, Ridge Regression, and Random Forest models.

Skills

Technical Skills

- Programming Languages: Python, Java, JavaScript, TypeScript, C++, C#
- AI & Machine Learning: Supervised & unsupervised learning, deep learning, NLP, computer vision, model evaluation, feature engineering
- Generative AI & LLMs: LLMs, RAG, embeddings, semantic search, prompt engineering, AI agents, agentic workflows, NL2SQL
- Data & Databases: SQL, PostgreSQL, Supabase, SQL Server, data preprocessing, data analysis, statistics
- Frameworks & Tools: Scikit-learn, TensorFlow, PyTorch, NLTK, Next.js, FastAPI, Prisma, Git/GitHub, Docker, Power BI

Soft Skills

- Strong problem-solving and analytical abilities
- Ability to work independently and collaboratively in teams
- Effective communication for technical and non-technical audiences
- Quick learner with eagerness to adopt new technologies and AI concepts

Languages

- Arabic: Native (C2)
- English: Professional proficiency (C1)